

Banking Risk Customer Analytics

End-to-End Data Analytics Project

Project Report

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Technology Stack

- PostgreSQL
- SQL
- Python
- Pandas
- SQLAlchemy
- Power BI
- Git & GitHub

Abstract

The Banking Risk Customer Analytics project is a complete end-to-end data analytics solution developed to simulate real-world banking operations.

The project demonstrates database design, ETL processes, SQL analytics, Python-based data analysis, and interactive Power BI dashboards. It helps identify customer behavior, financial risk, loan performance, transaction trends, branch performance, and merchant activity.

The objective is to provide business stakeholders with actionable insights for informed decision-making.

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Introduction

Banks generate millions of records every day through customers, accounts, transactions, loans, and merchant activities.

Analyzing this data manually is inefficient.

This project provides an automated analytics solution capable of generating business reports and interactive dashboards for executive decision-making.

Business Problem

Banks face several operational challenges:

- Identifying high-risk customers
- Monitoring loan defaults
- Tracking branch performance
- Detecting fraudulent transactions
- Understanding customer behavior
- Measuring financial performance

Without a centralized reporting system, decision-making becomes slower and less effective.

Project Objectives

- Design a normalized PostgreSQL database
- Generate realistic banking datasets
- Build a robust ETL pipeline using Python
- Perform SQL analysis from basic to advanced
- Create analytical reports using Python
- Develop interactive Power BI dashboards
- Document the complete project professionally

Technology Stack

Technology	Purpose
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PostgreSQL	Database

SQL	Data Analysis
Python	ETL & Analytics
Pandas	Data Processing
SQLAlchemy	Database Connectivity
Power BI	Visualization
GitHub	Version Control

Database Design

The database contains:

- Customers
- Accounts
- Transactions
- Loans
- Branches
- Merchants
- Loan Types
- Account Types

Relationships were implemented using primary keys and foreign keys to maintain referential integrity.

ETL Process

The ETL pipeline follows these stages:

1. Extract raw datasets
2. Clean and validate data
3. Transform columns and data types
4. Load data into PostgreSQL
5. Verify record counts
6. Generate validation reports

SQL Analysis

The project includes SQL queries covering:

- Basic SQL
- Intermediate SQL
- Advanced SQL
- Window Functions
- Common Table Expressions (CTEs)
- Recursive CTEs
- Views
- Materialized Views

- Functions
- Procedures
- Triggers
- Transactions
- Indexing
- Query Optimization

Python Analytics

Separate analysis modules were created for:

- Customer Analysis
- Account Analysis
- Transaction Analysis
- Loan Analysis
- Branch Analysis
- Merchant Analysis
- Risk Analysis
- Executive Summary

Each module connects directly to PostgreSQL using SQLAlchemy and generates analytical reports.

Power BI Dashboard

The dashboard includes:

- Executive Dashboard
- Customer Dashboard
- Account Dashboard
- Transaction Dashboard
- Loan Dashboard
- Branch Dashboard
- Merchant Dashboard
- Risk Dashboard

Key visualizations include KPI cards, charts, slicers, drill-through pages, and business insights.

Business Insights

Some key findings include:

- Customer distribution across states and cities
- Account balance trends
- Monthly transaction volumes
- Loan portfolio distribution
- Branch performance rankings

- Merchant transaction analysis
- High-risk customer identification
- Fraud indicators

These insights support strategic decision-making for bank management.

Challenges Faced

During development, the following challenges were encountered:

- Database schema design
- Foreign key constraint issues
- Data validation errors
- Duplicate records
- Missing values
- SQL query optimization
- Power BI relationship modeling
- ETL debugging

Each issue was resolved through testing, validation, and optimization.

Future Enhancements

Potential improvements include:

- Real-time data streaming
- Machine Learning for fraud detection
- Credit risk prediction models
- Customer churn prediction
- Banking recommendation engine
- Cloud deployment (AWS/Azure)
- REST API integration
- Automated dashboard refresh

Conclusion

The Banking Risk Customer Analytics project demonstrates an end-to-end analytics workflow from data generation to executive reporting.

It combines database management, SQL, Python, ETL, and Power BI into a unified solution that reflects real-world enterprise analytics practices.

This project serves as a strong portfolio piece for Data Analyst, BI Developer, SQL Developer, and Data Engineer roles.

References

- PostgreSQL Documentation
- Microsoft Power BI Documentation
- Python Documentation
- Pandas Documentation
- SQLAlchemy Documentation
- Git Documentation